Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Pareira Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
2487 W Turlock RD	Snelling	Merced	95369

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1947

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X042-X200-X054-XXXX X042-X200-X055-XXXX X042-X200-X057-XXXX X042-X240-X006-XXXX X042-X240-X039-XXXX

B. OPERATORS

Pareira, Justin

Operator name: Pareira, Justin

Telephone no.: (209) 722-5876 (Landline) (209) 777-1225 (Cellular)

Mailing Address Number and Street	City	State	Zip Code
2487 W Turlock RD	Snelling	CA	95369

This operator is responsible for paying permit fees.

C. OWNERS

Pareira, Leonard

Legal owner name: Pareira, Leonard

Telephone no.: (209) 722-5876 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
2487 W Turlock RD	Snelling	CA	95369

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	180	300	100	100
Number under roof	600	100	0	0	0	0
Maximum number	600	107	180	300	100	100
Average number	575	100	180	300	100	100
Avg live weight (lbs)	1,200	1,300	900	650		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 70 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 21,457.59 *tons per reporting period*

Total nitrogen from manure: 266,865.56 *lbs per reporting period*

After ammonia losses (30% loss applied): 186,805.89 *lbs per reporting period*

Total phosphorus from manure: 43,487.75 *lbs per reporting period*

Total potassium from manure: 110,117.81 *lbs per reporting period*

Total salt from manure: 293,733.75 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 19,682,000 *gallons*

Total nitrogen generated: 115,313.01 *lbs*

Total phosphorus generated: 12,176.43 *lbs*

Total potassium generated: 171,711.96 *lbs*

Total salt generated: 1,017,224.79 *lbs*

	19,682,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	19,682,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Merced River	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/30/2023	Solid commercial fertilizer *UN 32*	24.39 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	15,593.99	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	15,593.99	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	21	21	2	process wastewater	X042-X200-X055-XXXX
10	6	6	2	process wastewater	X042-X240-X006-XXXX X042-X240-X038-XXXX X042-X240-X039-XXXX
2	5	5	2	process wastewater	X042-X200-X055-XXXX
3	56	56	2	process wastewater	X042-X200-X055-XXXX X042-X240-X039-XXXX
4	23	23	2	process wastewater	X042-X200-X055-XXXX
5	37	37	2	process wastewater	X042-X200-X055-XXXX X042-X240-X039-XXXX
6	39	39	2	process wastewater	X042-X240-X039-XXXX
7	37	37	2	process wastewater	X042-X240-X039-XXXX
8	13	13	2	process wastewater	X042-X240-X006-XXXX X042-X240-X039-XXXX
9	22	22	2	process wastewater	X042-X240-X006-XXXX X042-X240-X039-XXXX
Almonds	83	83	1	manure	X042-X280-X023-XXXX
L-1	24	24	2	manure	X042-X200-X056-XXXX
L-2	3	3	2	manure	X042-X200-X056-XXXX
L-3	21	21	2	manure	X042-X200-X056-XXXX
Totals for areas that were used for application	390	390	27		
Totals for areas that were not used for application					
Land application area totals	390	390	27		

B. CROPS AND HARVESTS

1

Field name: 1

1

11/06/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 21 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	300.00 *ton*	As-is		61.6	5,920.00	1,310.00	8,520.00		7.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.29	169.14	37.43	243.43	822.86

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 21 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	620.00 *ton*	As-is		69.6	2,920.00	740.00	4,760.00		5.41

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	29.52	172.42	43.70	281.07	971.12

10

Field name: 10

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 6 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	90.00 *ton*	As-is		65.0	5,750.00	1,140.00	7,640.00		8.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.00	172.50	34.20	229.20	840.00

10

06/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 6 Plant date: 06/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/18/2023	140.00 *ton*	As-is		60.7	4,400.00	1,190.00	7,540.00		6.73

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	23.33	205.33	55.53	351.87	1,234.28

2

Field name: 2

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 5 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/30/2023	75.00 *ton*	As-is		65.8	5,780.00	1,020.00	7,910.00		8.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.00	173.40	30.60	237.30	831.06

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 5 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/18/2023	170.00 *ton*	As-is		71.0	2,760.00	770.00	4,850.00		5.59

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	34.00	187.68	52.36	329.80	1,102.35

3

Field name: 3

11/07/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 56 Plant date: 11/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	820.00 *ton*	As-is		63.8	5,970.00	1,280.00	8,930.00		8.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.64	174.84	37.49	261.52	858.72

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 56 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	1,850.00 *ton*	As-is		71.2	3,020.00	800.00	4,850.00		5.39

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	33.04	199.54	52.86	320.45	1,025.64

4

Field name: 4

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 23 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/30/2023	310.00 *ton*	As-is		64.8	6,140.00	1,430.00	10,300.00		8.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	13.48	165.51	38.55	277.65	797.05

4

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 23 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/18/2023	750.00 *ton*	As-is		68.1	3,090.00	1,040.00	5,620.00		4.79

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	32.61	201.52	67.83	366.52	996.53

5

Field name: 5

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/30/2023	620.00 *ton*	As-is		70.0	5,340.00	1,040.00	7,110.00		7.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	16.76	178.96	34.85	238.28	794.27

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	1,050.00 *ton*	As-is		70.3	3,920.00	870.00	4,890.00		6.11

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	28.38	222.49	49.38	277.54	1,029.95

6

Field name: 6

11/07/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 39 Plant date: 11/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	760.00 *ton*	As-is		72.8	4,470.00	980.00	6,800.00		7.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	19.49	174.22	38.19	265.03	837.48

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 39 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	1,150.00 *ton*	As-is		65.2	3,060.00	1,110.00	6,030.00		5.92

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	29.49	180.46	65.46	355.62	1,214.97

7

Field name: 7

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/30/2023	620.00 *ton*	As-is		70.8	4,840.00	930.00	5,960.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	16.76	162.21	31.17	199.74	645.87

7

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	940.00 *ton*	As-is		64.5	3,660.00	920.00	5,280.00		5.01

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	25.41	185.97	46.75	268.28	903.70

8

Field name: 8

11/07/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 13 Plant date: 11/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	195.00 *ton*	As-is		70.2	4,950.00	1,050.00	7,210.00		7.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.00	148.50	31.50	216.30	706.26

06/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 13 Plant date: 06/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/18/2023	370.00 *ton*	As-is		67.7	3,070.00	860.00	5,080.00		3.49

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	28.46	174.75	48.95	289.17	641.68

9

Field name: 9

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 22 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	360.00 *ton*	As-is		67.4	5,810.00	1,220.00	8,320.00		7.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	16.36	190.15	39.93	272.29	832.19

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 22 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	630.00 *ton*	As-is		66.4	3,060.00	920.00	5,630.00		5.19

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	28.64	175.25	52.69	322.45	998.74

Almonds

Field name: Almonds

10/24/2018: Almond Field Run

Crop: Almond Field Run Acres planted: 83 Plant date: 10/24/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	350.00 *ton*	As-is		10.8	16,700.00	1,170.00	13,600.00		16.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.50	193.50	32.85	209.25	0.00
Total actual harvest content	4.22	140.84	9.87	114.70	1,263.85

L-1

Field name: L-1

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 24 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	480.00 *ton*	As-is		70.4	4,580.00	1,080.00	7,500.00		9.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	20.00	183.20	43.20	300.00	1,089.28

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 24 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	550.00 *ton*	As-is		60.5	4,400.00	1,300.00	7,700.00		6.29

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	22.92	201.67	59.58	352.92	1,138.75

L-2

Field name: L-2

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 3 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	40.00 *ton*	As-is		62.0	6,400.00	1,370.00	9,300.00		7.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	13.33	170.67	36.53	248.00	729.60

L-2

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 3 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	70.00 *ton*	As-is		60.5	4,420.00	1,250.00	7,670.00		6.29

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	23.33	206.27	58.33	357.93	1,159.46

L-3

Field name: L-3

11/06/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 21 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	400.00 *ton*	As-is		70.6	4,220.00	880.00	5,730.00		7.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	19.05	160.76	33.52	218.29	784.00

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 21 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	672.00 *ton*	As-is		70.4	2,930.00	770.00	5,210.00		5.31

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	32.00	187.52	49.28	333.44	1,005.93

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 11/06/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 11/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	96.40	9.44	176.25	865.21	336,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	52.57	3,780,000.00 *gal*
Application event totals		96.40	9.44	176.25	917.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	56.42	11.67	95.68	604.27	288,000.00 *gal*
Application event totals		56.42	11.67	95.68	604.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	39.05	3,780,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.36	11.71	111.49	950.66	336,000.00 *gal*
Application event totals		64.36	11.71	111.49	950.66	

1 - 06/03/2023: Corn, silage

Field name: 1

Crop: Corn, silage Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

1 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	44.63	4,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	44.63	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	39.05	3,780,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.05	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.03	6.93	141.53	873.22	336,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	39.05	3,780,000.00 *gal*
Application event totals		71.03	6.93	141.53	912.28	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.96	6.43	131.42	810.85	312,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	36.26	3,510,000.00 *gal*
Application event totals		65.96	6.43	131.42	847.11	

1 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.96	6.43	131.42	810.85	312,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	36.26	3,510,000.00 *gal*
Application event totals		65.96	6.43	131.42	847.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	33.48	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	33.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	33.48	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	33.48	

10 - 11/05/2022: Oats, silage-soft dough

Field name: 10

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	96.40	9.44	176.25	865.21	96,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	39.43	810,000.00 *gal*
Application event totals		96.40	9.44	176.25	904.64	

10 - 11/05/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.83	13.62	111.62	704.99	96,000.00 *gal*
Application event totals		65.83	13.62	111.62	704.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.36	11.71	111.49	950.66	96,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	29.29	810,000.00 *gal*
Application event totals		64.36	11.71	111.49	979.95	

10 - 06/04/2023: Corn, silage

Field name: 10

Crop: Corn, silage

Plant date: 06/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	48.82	1,350,000.00 *gal*
Application event totals		0.00	0.00	0.00	48.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	39.05	1,080,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

10 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.03	6.93	141.53	873.22	96,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	39.05	1,080,000.00 *gal*
Application event totals		71.03	6.93	141.53	912.28	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.03	6.93	141.53	873.22	96,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	39.05	1,080,000.00 *gal*
Application event totals		71.03	6.93	141.53	912.28	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.03	6.93	141.53	873.22	96,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	39.05	1,080,000.00 *gal*
Application event totals		71.03	6.93	141.53	912.28	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	34.17	945,000.00 *gal*
Application event totals		0.00	0.00	0.00	34.17	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	29.29	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

2 - 11/04/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough Plant date: 11/04/2022

2 - 11/04/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	115.68	11.33	211.50	1,038.25	96,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	47.32	810,000.00 *gal*
Application event totals		115.68	11.33	211.50	1,085.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.50	8.17	66.97	422.99	48,000.00 *gal*
Application event totals		39.50	8.17	66.97	422.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	77.23	14.05	133.79	1,140.79	96,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	35.15	810,000.00 *gal*
Application event totals		77.23	14.05	133.79	1,175.94	

2 - 06/11/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 06/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	228.32	6.52	92.75	164.90	47,500.00 *gal*
Application event totals		228.32	6.52	92.75	164.90	

2 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	46.87	1,080,000.00 *gal*
Application event totals		0.00	0.00	0.00	46.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	35.15	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	29.29	675,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	29.29	675,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	23.43	540,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.43	

2 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	23.43	540,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	23.43	540,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.43	

3 - 11/07/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough

Plant date: 11/07/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	123.94	12.14	226.60	1,112.41	1,152,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	50.70	9,720,000.00 *gal*
Application event totals		123.94	12.14	226.60	1,163.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.32	8.76	71.76	453.21	576,000.00 *gal*
Application event totals		42.32	8.76	71.76	453.21	

3 - 11/07/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.95	12.55	119.45	1,018.57	960,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	31.38	8,100,000.00 *gal*
Application event totals		68.95	12.55	119.45	1,049.95	

3 - 06/03/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	48.12	12,420,000.00 *gal*
Application event totals		0.00	0.00	0.00	48.12	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	39.75	10,260,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.75	

07/02/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

3 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.50	6.68	136.48	842.03	864,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	37.66	9,720,000.00 *gal*
Application event totals		68.50	6.68	136.48	879.69	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.50	6.68	136.48	842.03	864,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	37.66	9,720,000.00 *gal*
Application event totals		68.50	6.68	136.48	879.69	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.69	6.31	128.89	795.25	816,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	35.57	9,180,000.00 *gal*
Application event totals		64.69	6.31	128.89	830.82	

08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	37.66	9,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	37.66	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	35.57	9,180,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.57	

4 - 11/04/2022: Oats, silage-soft dough

Field name: 4

Crop: Oats, silage-soft dough Plant date: 11/04/2022

4 - 11/04/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	100.59	9.85	183.91	902.83	384,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	41.14	3,240,000.00 *gal*
Application event totals		100.59	9.85	183.91	943.97	

03/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.39	13.32	109.20	689.66	360,000.00 *gal*
Application event totals		64.39	13.32	109.20	689.66	

04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.15	12.22	116.34	992.00	384,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	30.56	3,240,000.00 *gal*
Application event totals		67.15	12.22	116.34	1,022.56	

4 - 06/11/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 06/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	228.32	6.52	92.75	164.90	218,500.00 *gal*
Application event totals		228.32	6.52	92.75	164.90	

4 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	45.85	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	45.85	

07/08/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	38.21	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	38.21	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	35.66	3,780,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.66	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	35.66	3,780,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.66	

08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	30.56	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	30.56	

4 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	30.56	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	30.56	

09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	30.56	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	30.56	

5 - 11/04/2022: Oats, silage-soft dough

Field name: 5

Crop: Oats, silage-soft dough Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	117.24	11.48	214.35	1,052.28	720,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	47.96	6,075,000.00 *gal*
Application event totals		117.24	11.48	214.35	1,100.24	

03/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.70	8.83	72.40	457.29	384,000.00 *gal*
Application event totals		42.70	8.83	72.40	457.29	

5 - 11/04/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.84	12.34	117.52	1,002.05	624,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	30.87	5,265,000.00 *gal*
Application event totals		67.84	12.34	117.52	1,032.92	

5 - 06/11/2023: Corn, silage

Field name: 5

Crop: Corn, silage

Plant date: 06/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	228.32	6.52	92.75	164.90	351,500.00 *gal*
Application event totals		228.32	6.52	92.75	164.90	

05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	44.33	7,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	44.33	

07/08/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	38.00	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	38.00	

5 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	38.00	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	38.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	34.83	5,940,000.00 *gal*
Application event totals		0.00	0.00	0.00	34.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	31.67	5,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	31.67	5,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	31.67	5,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.67	

6 - 11/07/2022: Oats, silage-soft dough

Field name: 6

Crop: Oats, silage-soft dough

Plant date: 11/07/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

6 - 11/07/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/22/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	118.65	11.62	216.92	1,064.87	768,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	48.53	6,480,000.00 *gal*
Application event totals		118.65	11.62	216.92	1,113.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.63	10.48	85.86	542.30	480,000.00 *gal*
Application event totals		50.63	10.48	85.86	542.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.31	12.61	120.07	1,023.79	672,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	31.54	5,670,000.00 *gal*
Application event totals		69.31	12.61	120.07	1,055.33	

6 - 06/03/2023: Corn, silage

Field name: 6

Crop: Corn, silage

Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	48.07	8,640,000.00 *gal*
Application event totals		0.00	0.00	0.00	48.07	

6 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	42.06	7,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	42.06	

07/03/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.03	6.93	141.53	873.22	624,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	39.05	7,020,000.00 *gal*
Application event totals		71.03	6.93	141.53	912.28	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.57	6.40	130.64	806.05	576,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	36.05	6,480,000.00 *gal*
Application event totals		65.57	6.40	130.64	842.10	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.57	6.40	130.64	806.05	576,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	36.05	6,480,000.00 *gal*
Application event totals		65.57	6.40	130.64	842.10	

6 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	33.05	5,940,000.00 *gal*
Application event totals		0.00	0.00	0.00	33.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	33.05	5,940,000.00 *gal*
Application event totals		0.00	0.00	0.00	33.05	

7 - 11/04/2022: Oats, silage-soft dough

Field name: 7

Crop: Oats, silage-soft dough Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	109.43	10.72	200.06	982.13	672,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	44.76	5,670,000.00 *gal*
Application event totals		109.43	10.72	200.06	1,026.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.70	8.83	72.40	457.29	384,000.00 *gal*
Application event totals		42.70	8.83	72.40	457.29	

7 - 11/04/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	73.05	13.29	126.56	1,079.13	672,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	33.25	5,670,000.00 *gal*
Application event totals		73.05	13.29	126.56	1,112.38	

7 - 06/12/2023: Corn, silage

Field name: 7

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	228.32	6.52	92.75	164.90	351,500.00 *gal*
Application event totals		228.32	6.52	92.75	164.90	

05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	44.33	7,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	44.33	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	38.00	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	38.00	

07/09/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

7 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	38.00	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	38.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	34.83	5,940,000.00 *gal*
Application event totals		0.00	0.00	0.00	34.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	31.67	5,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	33.25	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	33.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	31.67	5,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.67	

8 - 11/07/2022: Oats, silage-soft dough

Field name: 8

Crop: Oats, silage-soft dough

Plant date: 11/07/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

8 - 11/07/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/22/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	111.23	10.89	203.36	998.32	240,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	45.50	2,025,000.00 *gal*
Application event totals		111.23	10.89	203.36	1,043.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.57	9.43	77.28	488.07	144,000.00 *gal*
Application event totals		45.57	9.43	77.28	488.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	59.41	10.81	102.91	877.53	192,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	27.04	1,620,000.00 *gal*
Application event totals		59.41	10.81	102.91	904.57	

8 - 06/04/2023: Corn, silage

Field name: 8

Crop: Corn, silage

Plant date: 06/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	54.08	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	54.08	

8 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	45.06	2,700,000.00 *gal*
Application event totals		0.00	0.00	0.00	45.06	

07/03/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	81.96	8.00	163.31	1,007.56	240,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	45.06	2,700,000.00 *gal*
Application event totals		81.96	8.00	163.31	1,052.63	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.57	6.40	130.64	806.05	192,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	36.05	2,160,000.00 *gal*
Application event totals		65.57	6.40	130.64	842.10	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.57	6.40	130.64	806.05	192,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	36.05	2,160,000.00 *gal*
Application event totals		65.57	6.40	130.64	842.10	

8 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	36.05	2,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	27.04	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	27.04	

9 - 11/05/2022: Oats, silage-soft dough

Field name: 9

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	105.17	10.30	192.27	943.87	384,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	43.01	3,240,000.00 *gal*
Application event totals		105.17	10.30	192.27	986.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.86	11.14	91.33	576.81	288,000.00 *gal*
Application event totals		53.86	11.14	91.33	576.81	

9 - 11/05/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	70.21	12.77	121.62	1,037.09	384,000.00 *gal*
Merced River	Ground water	0.00	0.00	0.00	31.95	3,240,000.00 *gal*
Application event totals		70.21	12.77	121.62	1,069.04	

9 - 06/12/2023: Corn, silage

Field name: 9

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	228.32	6.52	92.75	164.90	209,000.00 *gal*
Application event totals		228.32	6.52	92.75	164.90	

05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	42.61	4,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	42.61	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	34.62	3,510,000.00 *gal*
Application event totals		0.00	0.00	0.00	34.62	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

9 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	34.62	3,510,000.00 *gal*
Application event totals		0.00	0.00	0.00	34.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	31.95	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	33.29	3,375,000.00 *gal*
Application event totals		0.00	0.00	0.00	33.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	31.95	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	31.95	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.95	

Almonds - 10/24/2018: Almond Field Run

Field name: Almonds

Crop: Almond Field Run

Plant date: 10/24/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Almonds - 10/24/2018: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	28.21	8,015,169.00 *gal*
Application event totals		0.00	0.00	0.00	28.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/16/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	98.31	32.43	190.57	1,063.63	400.00 *ton*
Application event totals		98.31	32.43	190.57	1,063.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	41.90	16,030,338.00 *gal*
Application event totals		0.00	0.00	0.00	41.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	25.60	0.00	0.00	0.00	
Application event totals		25.60	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	41.90	16,030,338.00 *gal*
Application event totals		0.00	0.00	0.00	41.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	25.60	0.00	0.00	0.00	
Application event totals		25.60	0.00	0.00	0.00	

Almonds - 10/24/2018: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	41.90	16,030,338.00 *gal*
Application event totals		0.00	0.00	0.00	41.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	41.90	16,030,338.00 *gal*
Application event totals		0.00	0.00	0.00	41.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	41.90	16,030,338.00 *gal*
Application event totals		0.00	0.00	0.00	41.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	20.95	8,015,169.00 *gal*
Application event totals		0.00	0.00	0.00	20.95	

L-1 - 11/05/2022: Oats, silage-soft dough

Field name: L-1

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	212.48	70.10	411.90	2,299.00	250.00 *ton*
Application event totals		212.48	70.10	411.90	2,299.00	

L-1 - 11/05/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	49.29	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	36.61	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.61	

L-1 - 06/02/2023: Corn, silage

Field name: L-1

Crop: Corn, silage

Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	192.78	151.47	63.11	5,060.48	270.00 *ton*
Application event totals		192.78	151.47	63.11	5,060.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	43.94	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	43.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

L-1 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	36.61	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	36.61	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	32.95	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	32.95	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	29.29	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

L-1 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	29.29	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

L-2 - 11/05/2022: Oats, silage-soft dough

Field name: L-2

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	203.98	67.30	395.43	2,207.04	30.00 *ton*
Application event totals		203.98	67.30	395.43	2,207.04	

10/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	39.43	405,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.43	

04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	29.29	405,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

L-2 - 06/02/2023: Corn, silage

Field name: L-2

Crop: Corn, silage

Plant date: 06/02/2023

L-2 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	199.92	157.08	65.45	5,247.90	35.00 *ton*
Application event totals		199.92	157.08	65.45	5,247.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	29.29	405,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	29.29	405,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	29.29	405,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

L-2 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	29.29	405,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	29.29	405,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	29.29	405,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	29.29	405,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

L-3 - 11/06/2022: Oats, silage-soft dough

Field name: L-3

Crop: Oats, silage-soft dough

Plant date: 11/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	213.70	70.50	414.26	2,312.14	220.00 *ton*
Application event totals		213.70	70.50	414.26	2,312.14	

L-3 - 11/06/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	50.70	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	50.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	37.66	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	37.66	

L-3 - 06/02/2023: Corn, silage

Field name: L-3

Crop: Corn, silage

Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	195.84	153.87	64.11	5,140.80	240.00 *ton*
Application event totals		195.84	153.87	64.11	5,140.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	41.84	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	41.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

L-3 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	35.57	3,442,500.00 *gal*
Application event totals		0.00	0.00	0.00	35.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	33.48	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	33.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	33.48	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	33.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	31.38	3,037,500.00 *gal*
Application event totals		0.00	0.00	0.00	31.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	30.34	2,936,250.00 *gal*
Application event totals		0.00	0.00	0.00	30.34	

L-3 - 06/02/2023: Corn, silage

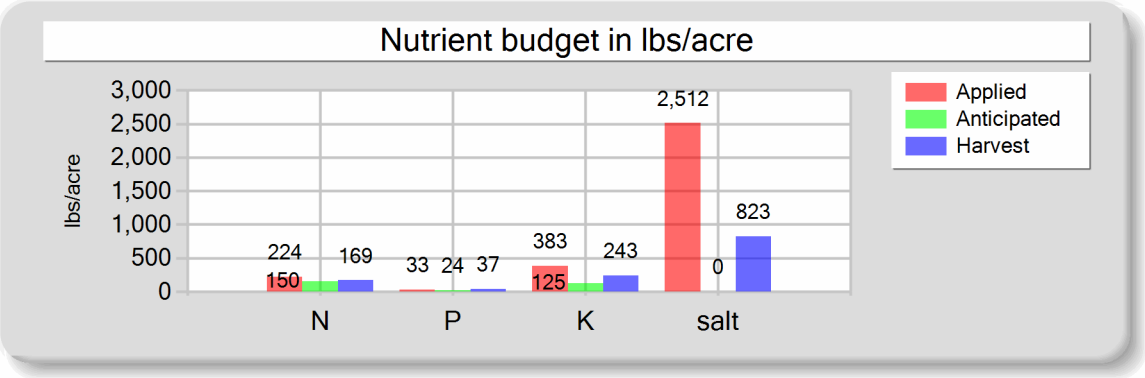
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Merced River	Ground water	0.00	0.00	0.00	29.29	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

B. NUTRIENT BUDGET

1 - 11/06/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	217.18	32.82	383.41	2,420.15
Fresh water	0.00	0.00	0.00	91.63
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	224.18	32.82	383.41	2,511.77
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	169.14	37.43	243.43	822.86
Nutrient balance	55.04	-4.61	139.98	1,688.92
Applied to removed ratio	1.33	0.88	1.58	3.05

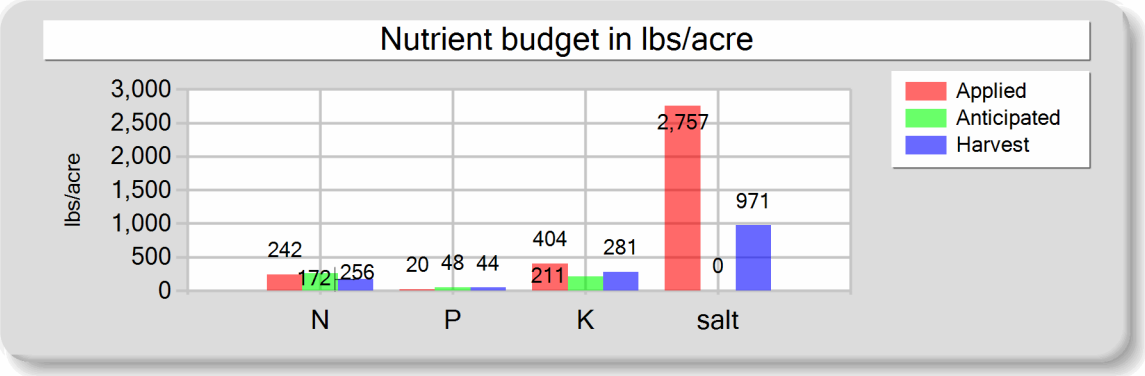
Fresh water applied	
7,560,000.00	*gallons*
278.41	*acre-inches*
13.26	*inches/acre*

Process wastewater applied	
960,000.00	*gallons*
35.35	*acre-inches*
1.68	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 06/03/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	202.95	19.80	404.37	2,494.92
Fresh water	0.00	0.00	0.00	262.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	241.95	19.80	404.37	2,757.14
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	172.42	43.70	281.07	971.12
Nutrient balance	69.53	-23.90	123.31	1,786.02
Applied to removed ratio	1.40	0.45	1.44	2.84

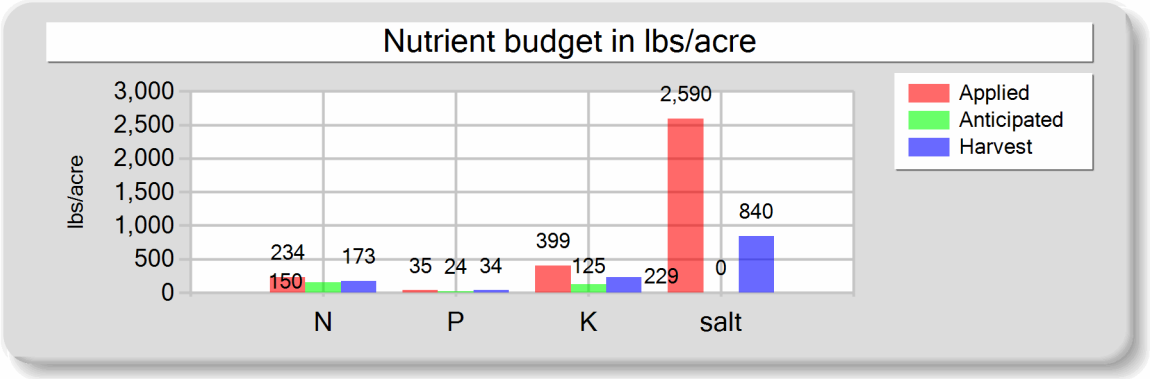
Fresh water applied	
25,380,000.00	*gallons*
934.66	*acre-inches*
44.51	*inches/acre*

Process wastewater applied	
960,000.00	*gallons*
35.35	*acre-inches*
1.68	*inches/acre*

Total harvests for the crop	
1	*harvests*

10 - 11/05/2022: Oats, silage-soft dough

Field name: 10 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	226.58	34.77	399.36	2,520.86
Fresh water	0.00	0.00	0.00	68.72
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	233.58	34.77	399.36	2,589.58
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	172.50	34.20	229.20	840.00
Nutrient balance	61.08	0.57	170.16	1,749.58
Applied to removed ratio	1.35	1.02	1.74	3.08

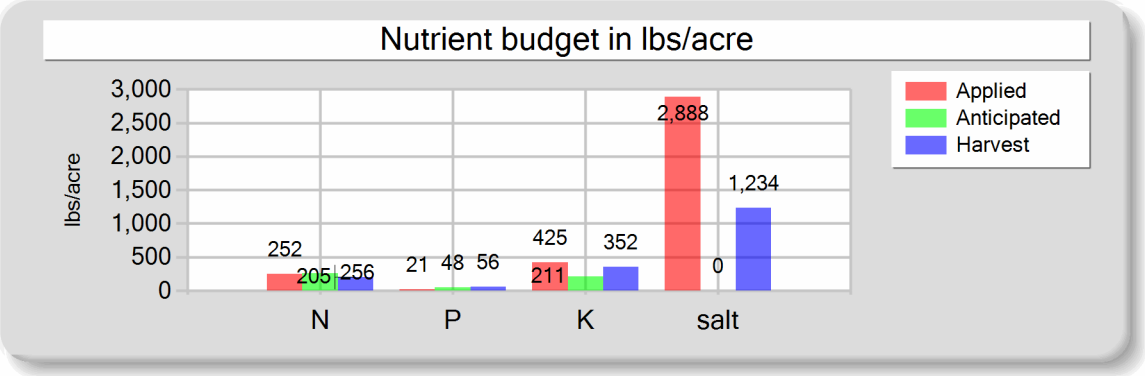
Fresh water applied	
1,620,000.00	*gallons*
59.66	*acre-inches*
9.94	*inches/acre*

Process wastewater applied	
288,000.00	*gallons*
10.61	*acre-inches*
1.77	*inches/acre*

Total harvests for the crop	
1	*harvests*

10 - 06/04/2023: Corn, silage

Field name: 10 Crop: Corn, silage Plant date: 06/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	213.10	20.79	424.59	2,619.66
Fresh water	0.00	0.00	0.00	268.50
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	252.10	20.79	424.59	2,888.16
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	205.33	55.53	351.87	1,234.28
Nutrient balance	46.76	-34.74	72.73	1,653.88
Applied to removed ratio	1.23	0.37	1.21	2.34

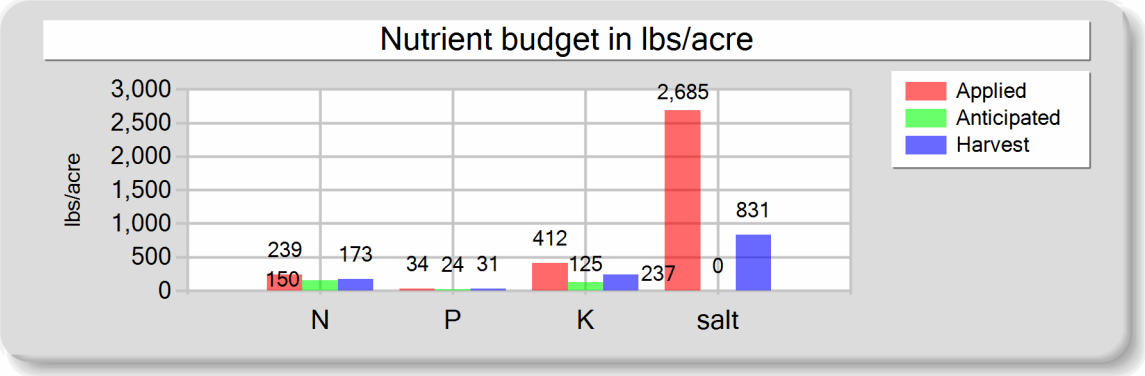
Fresh water applied	
7,425,000.00	*gallons*
273.44	*acre-inches*
45.57	*inches/acre*

Process wastewater applied	
288,000.00	*gallons*
10.61	*acre-inches*
1.77	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 11/04/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	232.40	33.55	412.26	2,602.04
Fresh water	0.00	0.00	0.00	82.47
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	239.40	33.55	412.26	2,684.50
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	173.40	30.60	237.30	831.06
Nutrient balance	66.00	2.95	174.96	1,853.44
Applied to removed ratio	1.38	1.10	1.74	3.23

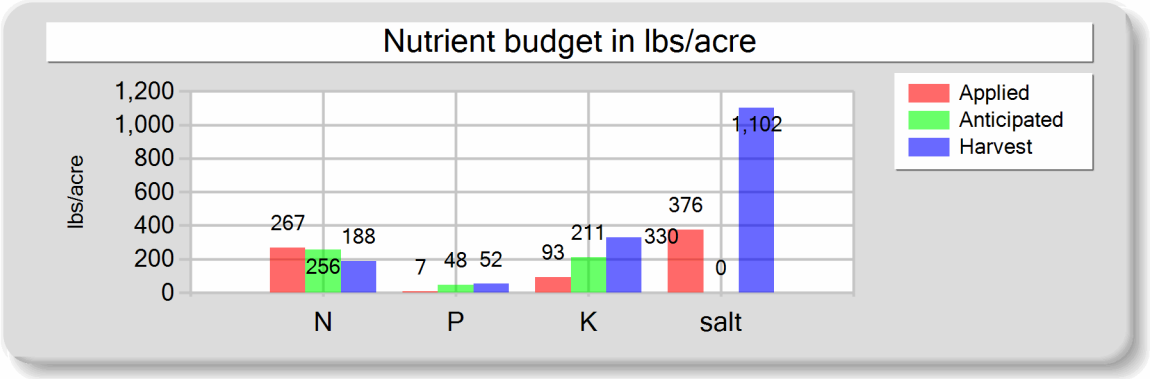
Fresh water applied	
1,620,000.00	*gallons*
59.66	*acre-inches*
11.93	*inches/acre*

Process wastewater applied	
240,000.00	*gallons*
8.84	*acre-inches*
1.77	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 06/11/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	228.32	6.52	92.75	164.90
Fresh water	0.00	0.00	0.00	210.89
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	267.32	6.52	92.75	375.79
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	187.68	52.36	329.80	1,102.35
Nutrient balance	79.64	-45.84	-237.05	-726.56
Applied to removed ratio	1.42	0.12	0.28	0.34

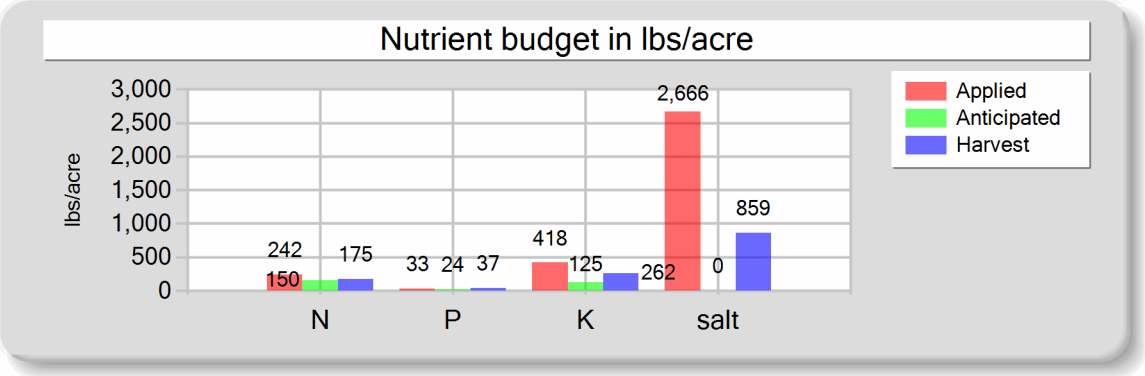
Fresh water applied	
4,860,000.00	*gallons*
178.98	*acre-inches*
35.80	*inches/acre*

Process wastewater applied	
47,500.00	*gallons*
1.75	*acre-inches*
0.35	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 11/07/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 11/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	235.21	33.44	417.81	2,584.18
Fresh water	0.00	0.00	0.00	82.08
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	242.21	33.44	417.81	2,666.26
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	174.84	37.49	261.52	858.72
Nutrient balance	67.38	-4.05	156.29	1,807.55
Applied to removed ratio	1.39	0.89	1.60	3.10

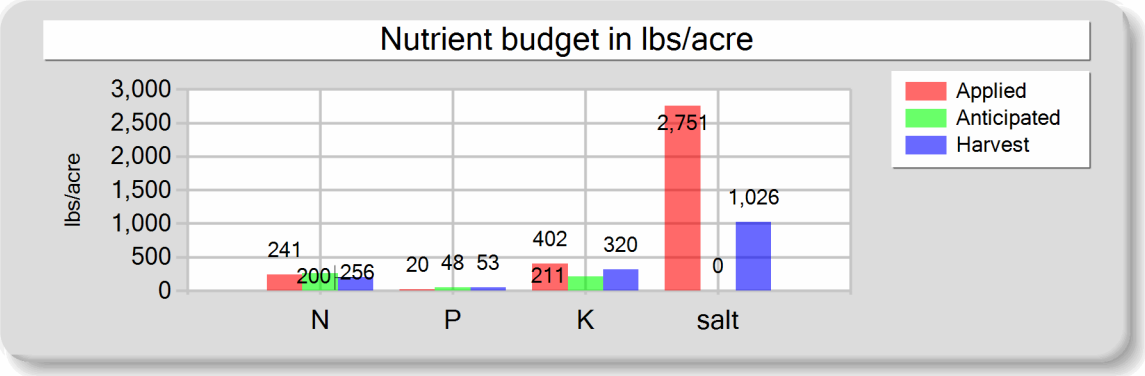
Fresh water applied	
17,820,000.00	*gallons*
656.25	*acre-inches*
11.72	*inches/acre*

Process wastewater applied	
2,688,000.00	*gallons*
98.99	*acre-inches*
1.77	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 06/03/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	201.68	19.68	401.85	2,479.32
Fresh water	0.00	0.00	0.00	271.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	240.68	19.68	401.85	2,751.31
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	199.54	52.86	320.45	1,025.64
Nutrient balance	41.15	-33.18	81.40	1,725.67
Applied to removed ratio	1.21	0.37	1.25	2.68

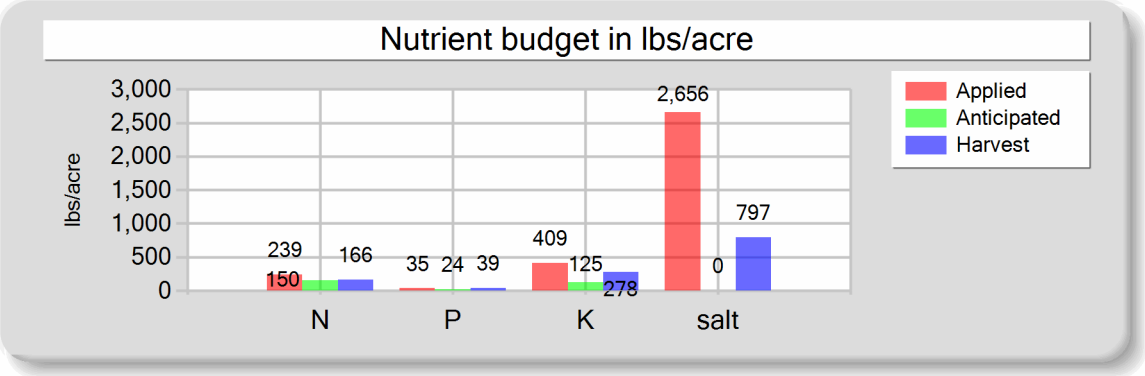
Fresh water applied	
70,200,000.00	*gallons*
2,585.23	*acre-inches*
46.16	*inches/acre*

Process wastewater applied	
2,544,000.00	*gallons*
93.69	*acre-inches*
1.67	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 11/04/2022: Oats, silage-soft dough

Field name: 4 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	232.14	35.39	409.44	2,584.48
Fresh water	0.00	0.00	0.00	71.71
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	239.14	35.39	409.44	2,656.19
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	165.51	38.55	277.65	797.05
Nutrient balance	73.63	-3.16	131.79	1,859.14
Applied to removed ratio	1.44	0.92	1.47	3.33

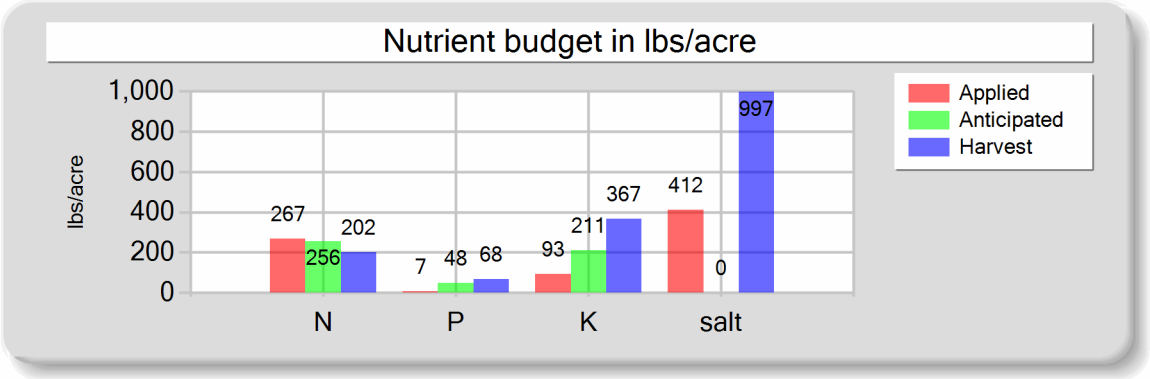
Fresh water applied	
6,480,000.00	*gallons*
238.64	*acre-inches*
10.38	*inches/acre*

Process wastewater applied	
1,128,000.00	*gallons*
41.54	*acre-inches*
1.81	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 06/11/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	228.32	6.52	92.75	164.90
Fresh water	0.00	0.00	0.00	247.06
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	267.32	6.52	92.75	411.96
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	201.52	67.83	366.52	996.53
Nutrient balance	65.80	-61.31	-273.77	-584.57
Applied to removed ratio	1.33	0.10	0.25	0.41

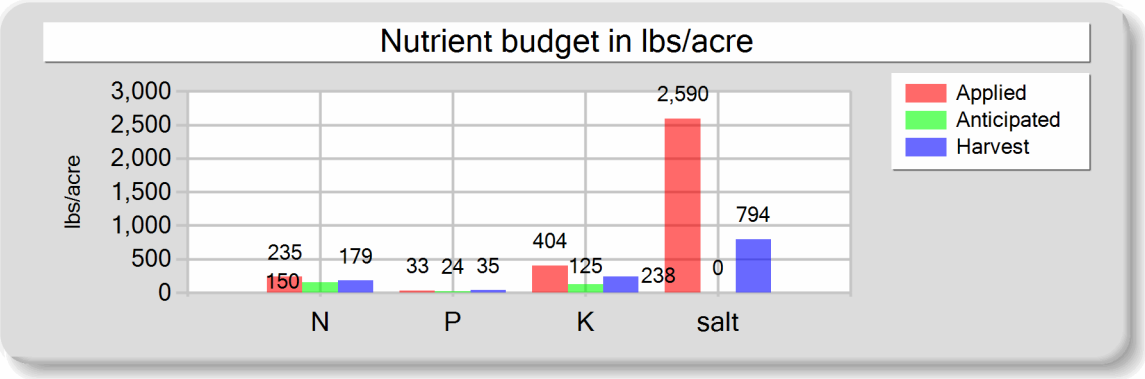
Fresh water applied	
26,190,000.00	*gallons*
964.49	*acre-inches*
41.93	*inches/acre*

Process wastewater applied	
218,500.00	*gallons*
8.05	*acre-inches*
0.35	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 11/04/2022: Oats, silage-soft dough

Field name: 5 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	227.78	32.66	404.27	2,511.62
Fresh water	0.00	0.00	0.00	78.83
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	234.78	32.66	404.27	2,590.45
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	178.96	34.85	238.28	794.27
Nutrient balance	55.82	-2.20	165.99	1,796.18
Applied to removed ratio	1.31	0.94	1.70	3.26

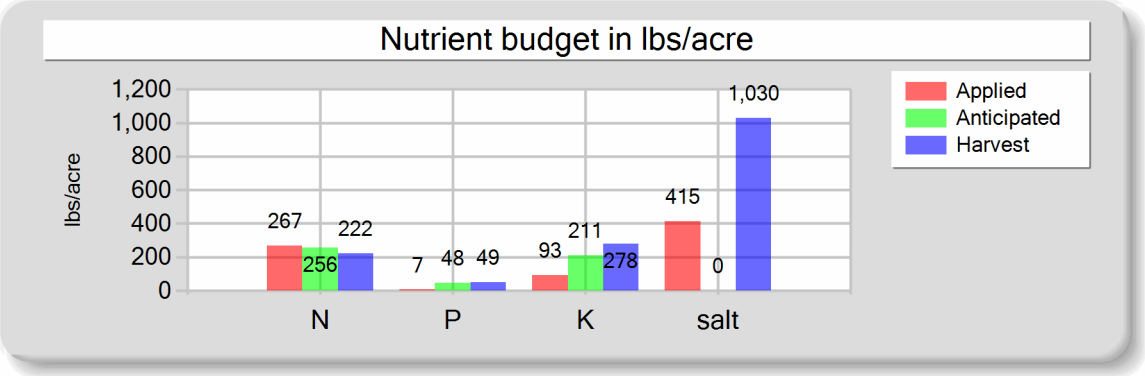
Fresh water applied	
11,340,000.00	*gallons*
417.61	*acre-inches*
11.29	*inches/acre*

Process wastewater applied	
1,728,000.00	*gallons*
63.64	*acre-inches*
1.72	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 06/11/2023: Corn, silage

Field name: 5 Crop: Corn, silage Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	228.32	6.52	92.75	164.90
Fresh water	0.00	0.00	0.00	250.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	267.32	6.52	92.75	415.06
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	222.49	49.38	277.54	1,029.95
Nutrient balance	44.83	-42.86	-184.79	-614.89
Applied to removed ratio	1.20	0.13	0.33	0.40

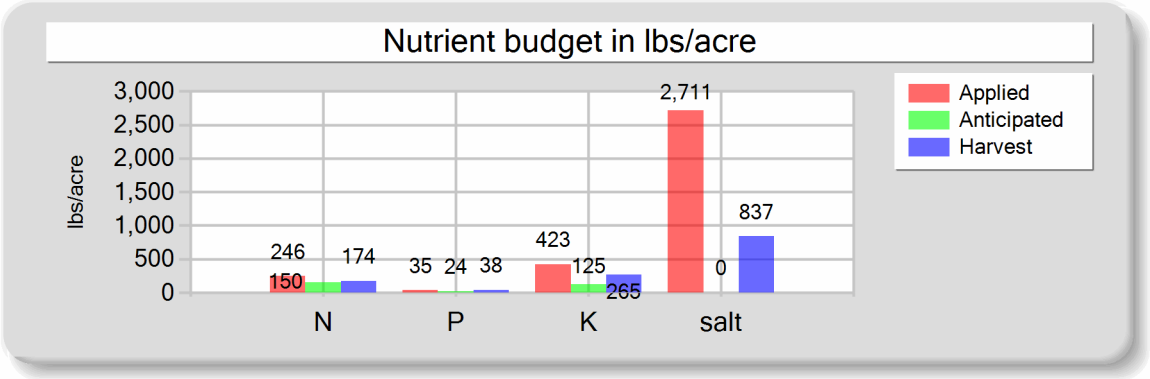
Fresh water applied	
42,660,000.00	*gallons*
1,571.02	*acre-inches*
42.46	*inches/acre*

Process wastewater applied	
351,500.00	*gallons*
12.94	*acre-inches*
0.35	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 11/07/2022: Oats, silage-soft dough

Field name: 6 Crop: Oats, silage-soft dough Plant date: 11/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	238.59	34.70	422.85	2,630.96
Fresh water	0.00	0.00	0.00	80.07
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	245.59	34.70	422.85	2,711.03
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	174.22	38.19	265.03	837.48
Nutrient balance	71.37	-3.49	157.82	1,873.55
Applied to removed ratio	1.41	0.91	1.60	3.24

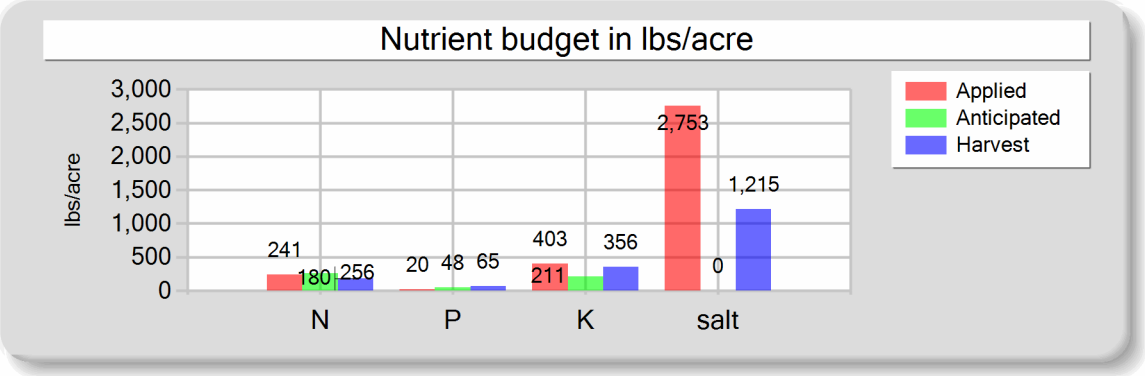
Fresh water applied	
12,150,000.00	*gallons*
447.44	*acre-inches*
11.47	*inches/acre*

Process wastewater applied	
1,920,000.00	*gallons*
70.71	*acre-inches*
1.81	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 06/03/2023: Corn, silage

Field name: 6 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	202.17	19.72	402.82	2,485.32
Fresh water	0.00	0.00	0.00	267.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	241.17	19.72	402.82	2,752.69
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	180.46	65.46	355.62	1,214.97
Nutrient balance	60.71	-45.74	47.20	1,537.73
Applied to removed ratio	1.34	0.30	1.13	2.27

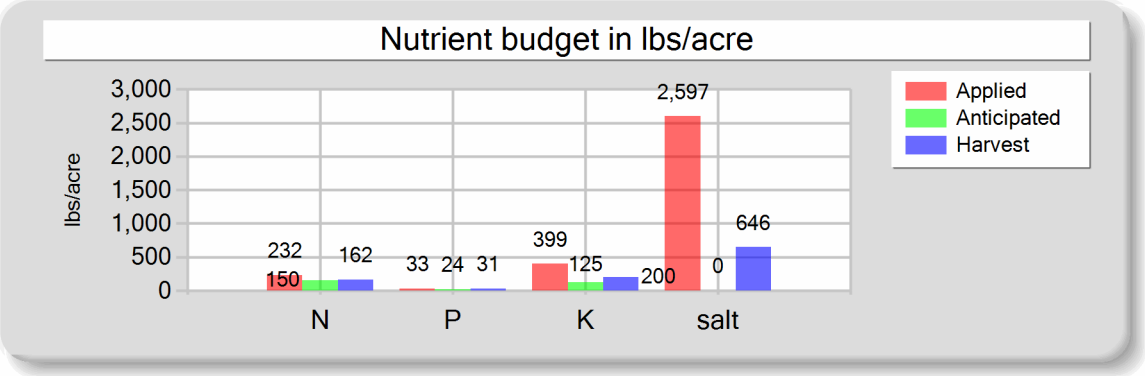
Fresh water applied	
48,060,000.00	*gallons*
1,769.89	*acre-inches*
45.38	*inches/acre*

Process wastewater applied	
1,776,000.00	*gallons*
65.40	*acre-inches*
1.68	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 11/04/2022: Oats, silage-soft dough

Field name: 7 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	225.18	32.84	399.02	2,518.55
Fresh water	0.00	0.00	0.00	78.01
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	232.18	32.84	399.02	2,596.56
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	162.21	31.17	199.74	645.87
Nutrient balance	69.97	1.67	199.28	1,950.68
Applied to removed ratio	1.43	1.05	2.00	4.02

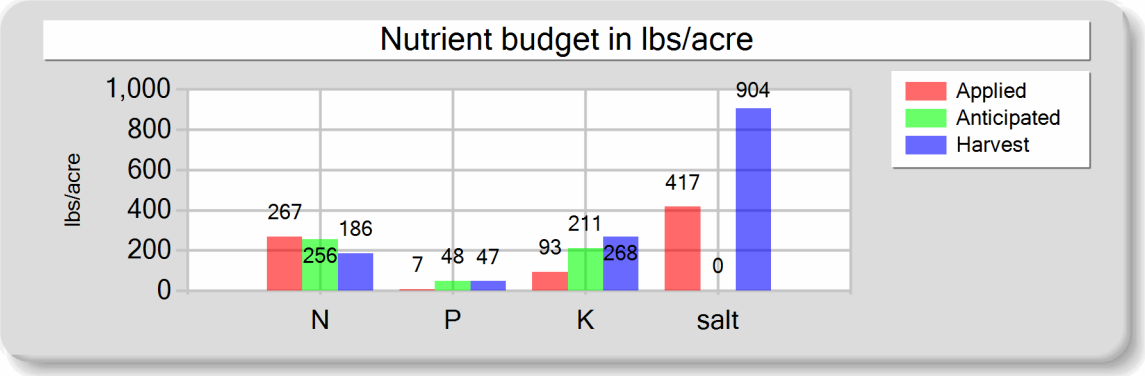
Fresh water applied	
11,340,000.00	*gallons*
417.61	*acre-inches*
11.29	*inches/acre*

Process wastewater applied	
1,728,000.00	*gallons*
63.64	*acre-inches*
1.72	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 06/12/2023: Corn, silage

Field name: 7 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	228.32	6.52	92.75	164.90
Fresh water	0.00	0.00	0.00	251.74
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	267.32	6.52	92.75	416.64
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	185.97	46.75	268.28	903.70
Nutrient balance	81.35	-40.23	-175.53	-487.05
Applied to removed ratio	1.44	0.14	0.35	0.46

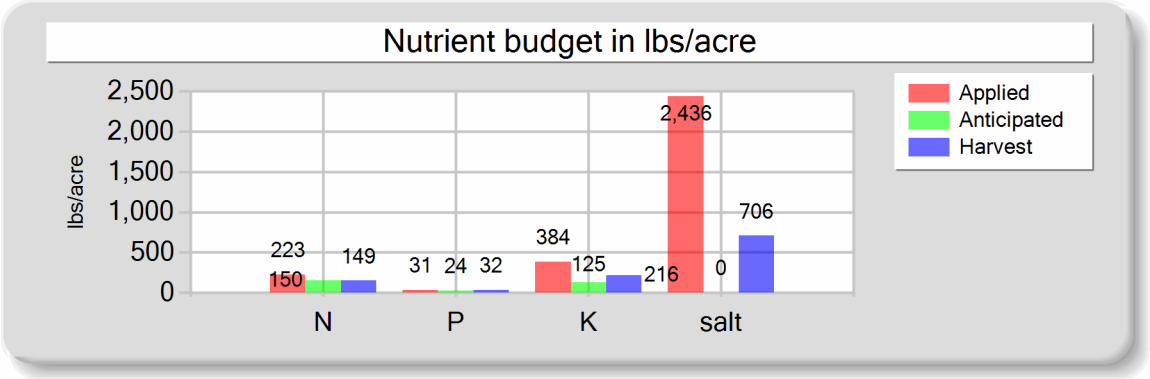
Fresh water applied	
42,930,000.00	*gallons*
1,580.97	*acre-inches*
42.73	*inches/acre*

Process wastewater applied	
351,500.00	*gallons*
12.94	*acre-inches*
0.35	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 11/07/2022: Oats, silage-soft dough

Field name: 8 Crop: Oats, silage-soft dough Plant date: 11/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	216.21	31.13	383.55	2,363.92
Fresh water	0.00	0.00	0.00	72.53
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	223.21	31.13	383.55	2,436.45
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	148.50	31.50	216.30	706.26
Nutrient balance	74.71	-0.37	167.25	1,730.19
Applied to removed ratio	1.50	0.99	1.77	3.45

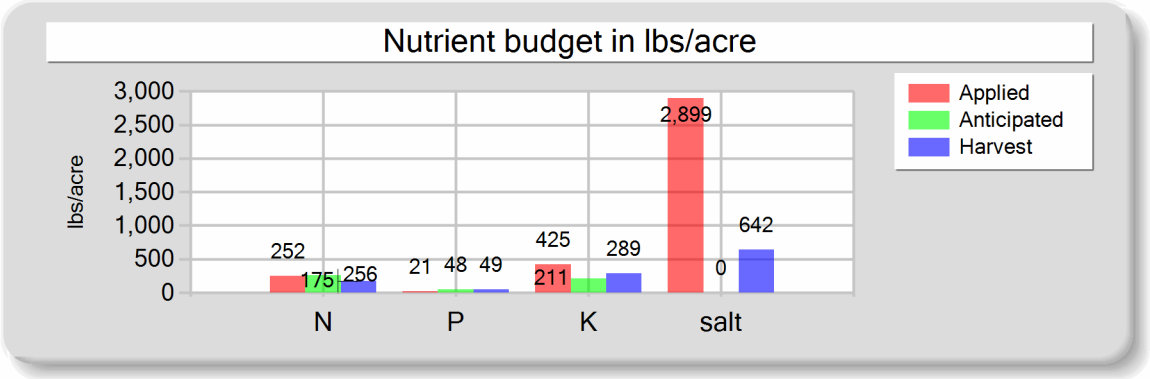
Fresh water applied	
3,645,000.00	*gallons*
134.23	*acre-inches*
10.33	*inches/acre*

Process wastewater applied	
576,000.00	*gallons*
21.21	*acre-inches*
1.63	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 06/04/2023: Corn, silage

Field name: 8 Crop: Corn, silage Plant date: 06/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	213.10	20.79	424.59	2,619.66
Fresh water	0.00	0.00	0.00	279.39
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	252.10	20.79	424.59	2,899.05
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	174.75	48.95	289.17	641.68
Nutrient balance	77.34	-28.16	135.42	2,257.38
Applied to removed ratio	1.44	0.42	1.47	4.52

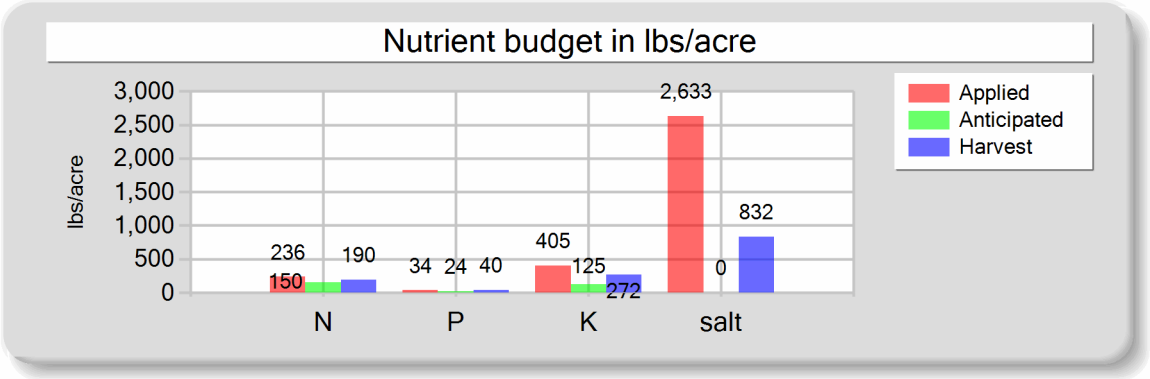
Fresh water applied	
16,740,000.00	*gallons*
616.48	*acre-inches*
47.42	*inches/acre*

Process wastewater applied	
624,000.00	*gallons*
22.98	*acre-inches*
1.77	*inches/acre*

Total harvests for the crop	
1	*harvests*

9 - 11/05/2022: Oats, silage-soft dough

Field name: 9 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	229.23	34.22	405.22	2,557.76
Fresh water	0.00	0.00	0.00	74.97
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	236.23	34.22	405.22	2,632.73
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	190.15	39.93	272.29	832.19
Nutrient balance	46.08	-5.71	132.93	1,800.54
Applied to removed ratio	1.24	0.86	1.49	3.16

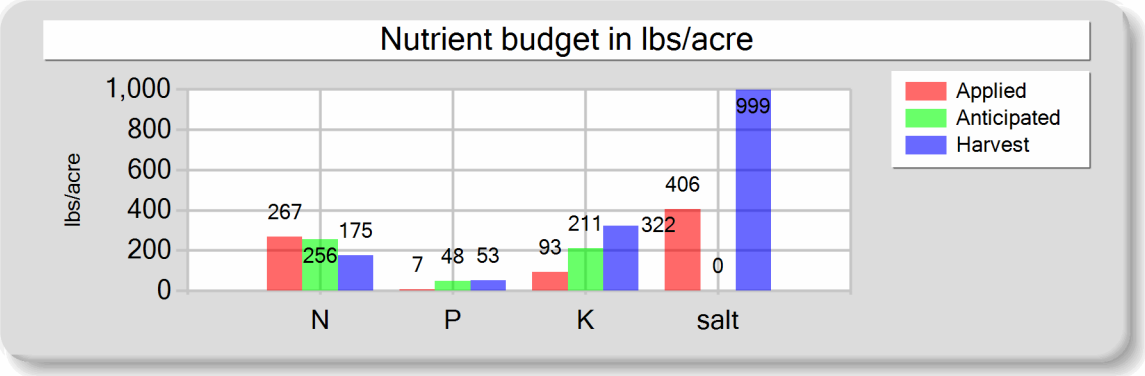
Fresh water applied	
6,480,000.00	*gallons*
238.64	*acre-inches*
10.85	*inches/acre*

Process wastewater applied	
1,056,000.00	*gallons*
38.89	*acre-inches*
1.77	*inches/acre*

Total harvests for the crop	
1	*harvests*

9 - 06/12/2023: Corn, silage

Field name: 9 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	228.32	6.52	92.75	164.90
Fresh water	0.00	0.00	0.00	240.98
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	267.32	6.52	92.75	405.88
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	175.25	52.69	322.45	998.74
Nutrient balance	92.06	-46.17	-229.69	-592.86
Applied to removed ratio	1.53	0.12	0.29	0.41

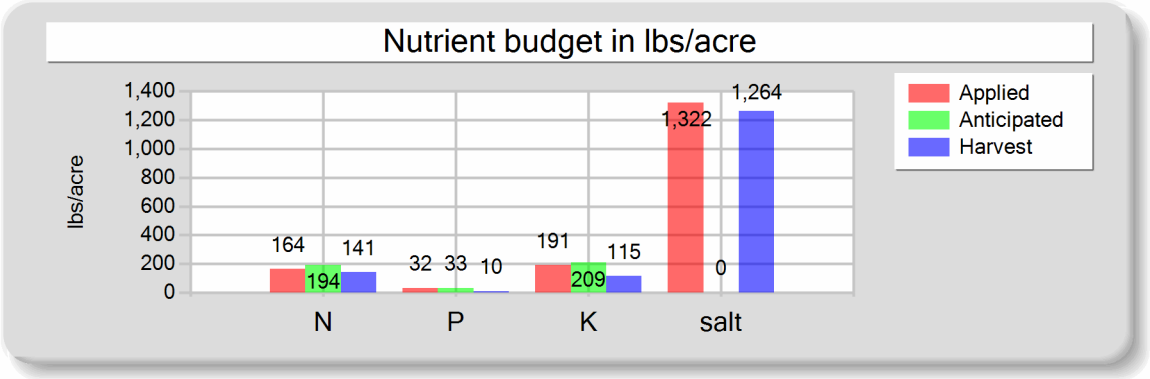
Fresh water applied	
24,435,000.00	*gallons*
899.86	*acre-inches*
40.90	*inches/acre*

Process wastewater applied	
209,000.00	*gallons*
7.70	*acre-inches*
0.35	*inches/acre*

Total harvests for the crop	
1	*harvests*

Almonds - 10/24/2018: Almond Field Run

Field name: Almonds Crop: Almond Field Run Plant date: 10/24/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	51.20	0.00	0.00	0.00
Dry manure	98.31	32.43	190.57	1,063.63
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	258.68
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	163.51	32.43	190.57	1,322.32
Anticipated crop nutrient removal	193.50	32.85	209.25	0.00
Actual crop nutrient removal	140.84	9.87	114.70	1,263.85
Nutrient balance	22.66	22.57	75.87	58.47
Applied to removed ratio	1.16	3.29	1.66	1.05

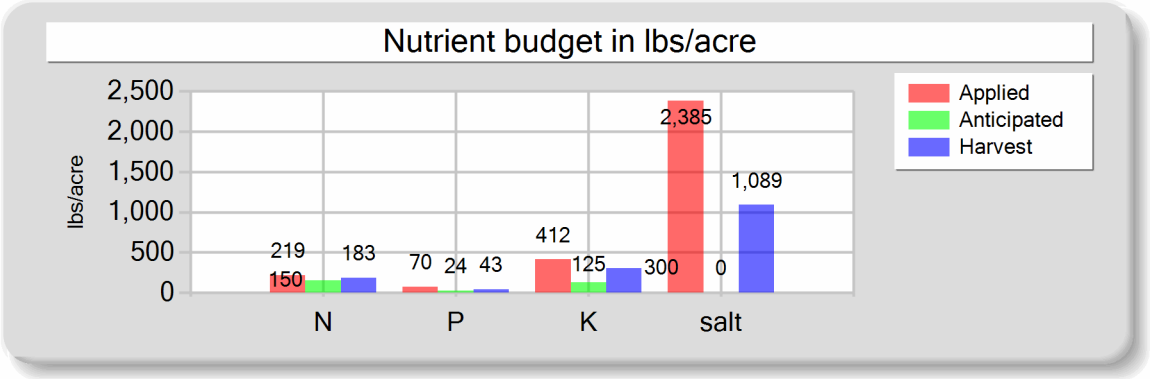
Fresh water applied	
96,182,028.00	*gallons*
3,542.06	*acre-inches*
42.68	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

L-1 - 11/05/2022: Oats, silage-soft dough

Field name: L-1 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	212.48	70.10	411.90	2,299.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	85.90
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	219.48	70.10	411.90	2,384.90
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	183.20	43.20	300.00	1,089.28
Nutrient balance	36.28	26.90	111.90	1,295.62
Applied to removed ratio	1.20	1.62	1.37	2.19

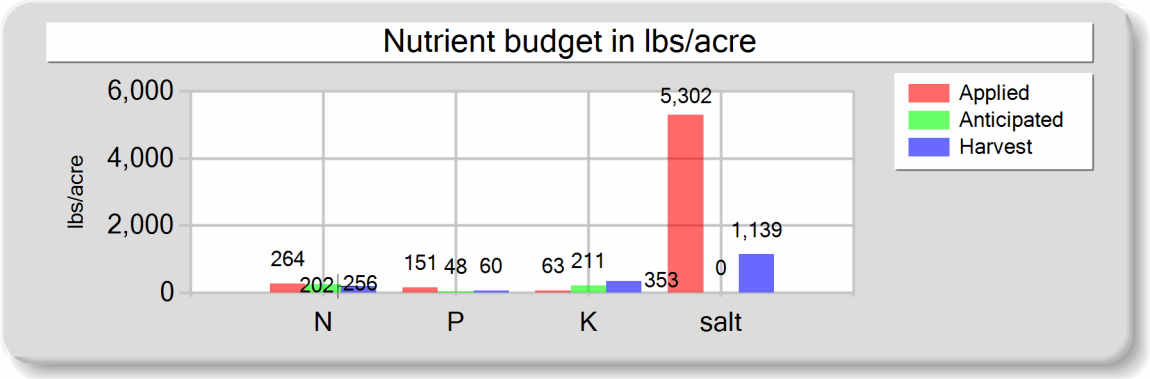
Fresh water applied	
8,100,000.00	*gallons*
298.30	*acre-inches*
12.43	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

L-1 - 06/02/2023: Corn, silage

Field name: L-1 Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	192.78	151.47	63.11	5,060.48
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	241.65
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	263.78	151.47	63.11	5,302.13
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	201.67	59.58	352.92	1,138.75
Nutrient balance	62.11	91.89	-289.80	4,163.37
Applied to removed ratio	1.31	2.54	0.18	4.66

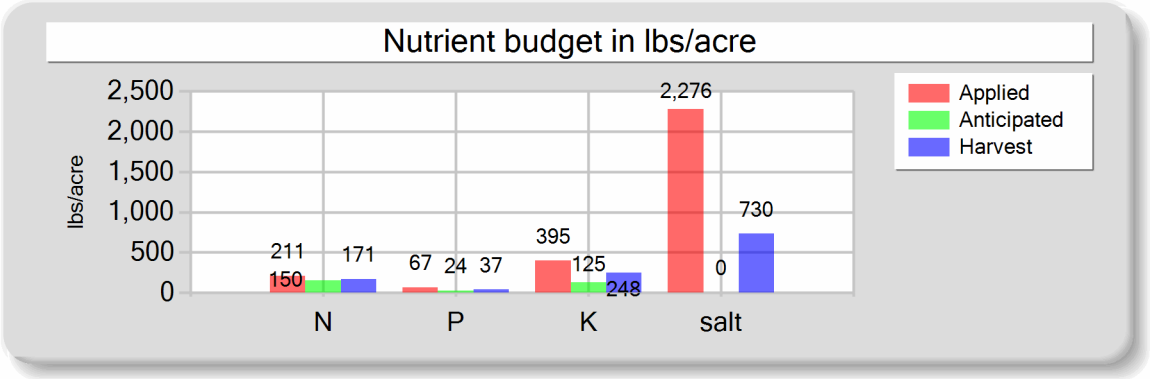
Fresh water applied
26,730,000.00 *gallons*
984.38 *acre-inches*
41.02 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

L-2 - 11/05/2022: Oats, silage-soft dough

Field name: L-2 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	203.98	67.30	395.43	2,207.04
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	68.72
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	210.98	67.30	395.43	2,275.76
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	170.67	36.53	248.00	729.60
Nutrient balance	40.32	30.76	147.43	1,546.16
Applied to removed ratio	1.24	1.84	1.59	3.12

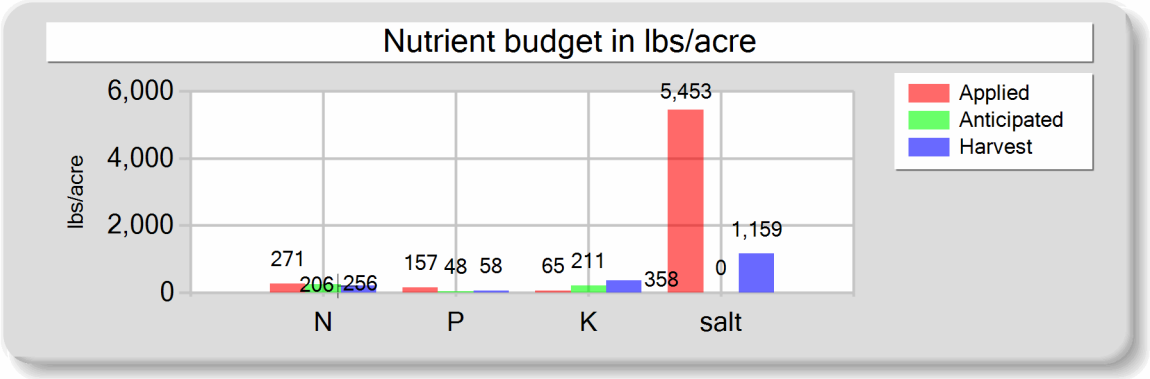
Fresh water applied	
810,000.00	*gallons*
29.83	*acre-inches*
9.94	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

L-2 - 06/02/2023: Corn, silage

Field name: L-2 Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	199.92	157.08	65.45	5,247.90
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	205.04
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	270.92	157.08	65.45	5,452.94
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	206.27	58.33	357.93	1,159.46
Nutrient balance	64.65	98.75	-292.48	4,293.48
Applied to removed ratio	1.31	2.69	0.18	4.70

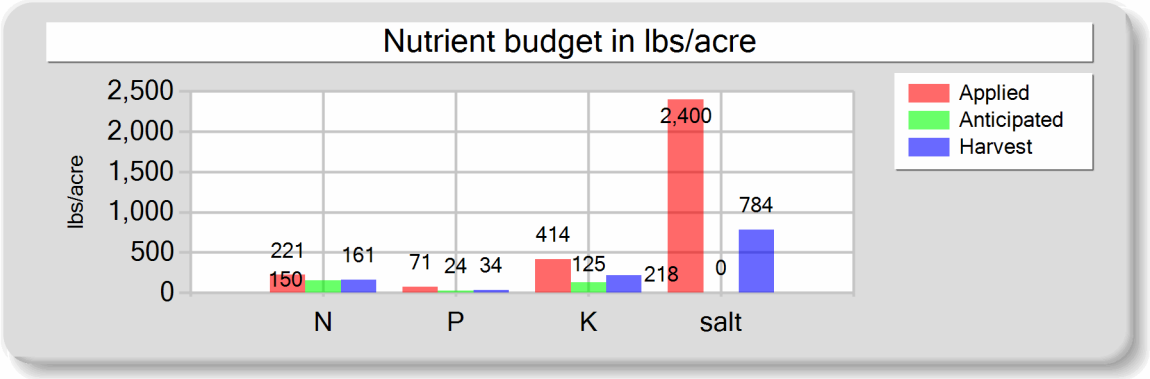
Fresh water applied	
2,835,000.00	*gallons*
104.40	*acre-inches*
34.80	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

L-3 - 11/06/2022: Oats, silage-soft dough

Field name: L-3 Crop: Oats, silage-soft dough Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	213.70	70.50	414.26	2,312.14
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	88.36
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	220.70	70.50	414.26	2,400.49
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	160.76	33.52	218.29	784.00
Nutrient balance	59.94	36.98	195.97	1,616.49
Applied to removed ratio	1.37	2.10	1.90	3.06

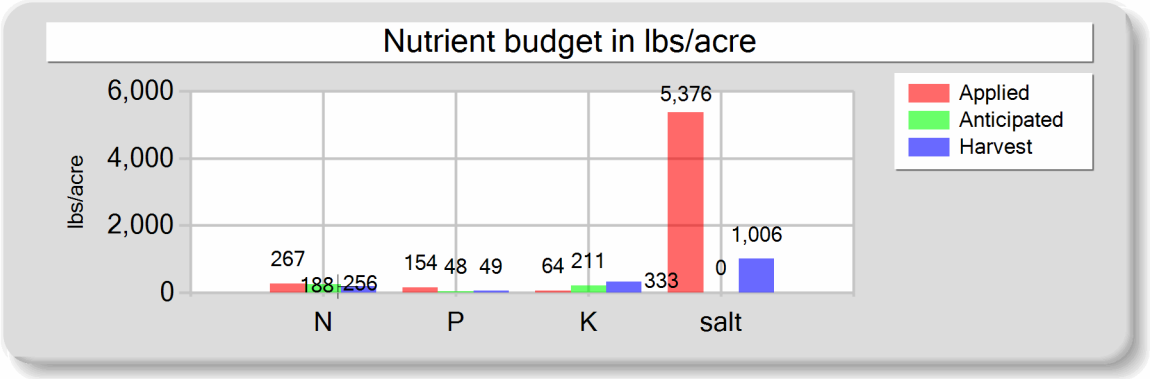
Fresh water applied	
7,290,000.00	*gallons*
268.47	*acre-inches*
12.78	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

L-3 - 06/02/2023: Corn, silage

Field name: L-3 Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	195.84	153.87	64.11	5,140.80
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	235.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	266.84	153.87	64.11	5,376.17
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	187.52	49.28	333.44	1,005.93
Nutrient balance	79.32	104.59	-269.33	4,370.25
Applied to removed ratio	1.42	3.12	0.19	5.34

Fresh water applied	
22,781,250.00	*gallons*
838.96	*acre-inches*
39.95	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/13/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 58.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,400.00	8,050.00	47,300.00							26.40
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/27/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 49.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,800.00	13,200.00	5,500.00							44.10
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/12/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 49.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,400.00	3,830.00	32,200.00							38.40
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

4QWW

Sample and source description: 4QWW

Sample date: 10/20/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	722.00	462.00	0.00	0.00	70.70	1,320.00								9,890.00	6,480
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 01/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	493.00	294.00	0.00	0.00	102.00	836.00								6,820.00	5,280
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Slurry

Sample and source description: Slurry

Sample date: 05/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2,880.00	728.00	0.00	0.00	82.20	1,170.00								6,840.00	2,080
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 06/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	482.00	300.00	0.00	0.00	87.70	835.00								6,570.00	7,120
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 07/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	532.00	356.00	0.00	0.00	51.90	1,060.00								7,580.00	6,540
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Slurry

Sample and source description: Slurry

Sample date: 10/19/2023 Material type: Process wastewater sludge Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,140.00	420.00	0.00	0.00	59.30	899.00								5,890.00	4,060
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4QWW

Sample and source description: 4QWW

Sample date: 10/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	619.00	353.00	0.00	0.00	56.80	1,160.00								8,450.00	6,300
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Merced River

Merced River

Fresh Water

Sample description: Fresh Water

Sample date: 10/06/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	35
DL	0.50	0.50	0.50								1.00	10

Fresh Water

Sample description: Fresh Water

Sample date: 04/05/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								30.90	26
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1 - 11/06/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,920.00	1,310.00	8,520.00		7.50
DL	100.00	100.00	100.00		0.01

1 - 06/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,920.00	740.00	4,760.00		5.41
DL	100.00	100.00	100.00		0.01

10 - 11/05/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,750.00	1,140.00	7,640.00		8.00
DL	100.00	100.00	100.00		0.01

10 - 06/04/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,400.00	1,190.00	7,540.00		6.73
DL	100.00	100.00	100.00		0.01

2 - 11/04/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,780.00	1,020.00	7,910.00		8.10
DL	100.00	100.00	100.00		0.01

2 - 06/11/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,760.00	770.00	4,850.00		5.59
DL	100.00	100.00	100.00		0.01

3 - 11/07/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,970.00	1,280.00	8,930.00		8.10
DL	100.00	100.00	100.00		0.01

3 - 11/07/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,720.00	1,310.00	9,300.00		8.60
DL	100.00	100.00	100.00		0.01

3 - 06/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,020.00	800.00	4,850.00		5.39
DL	100.00	100.00	100.00		0.01

4 - 11/04/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,140.00	1,430.00	10,300.00		8.40
DL	100.00	100.00	100.00		0.01

4 - 06/11/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,090.00	1,040.00	5,620.00		4.79
DL	100.00	100.00	100.00		0.01

5 - 11/04/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,340.00	1,040.00	7,110.00		7.90
DL	100.00	100.00	100.00		0.01

5 - 06/11/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,920.00	870.00	4,890.00		6.11
DL	100.00	100.00	100.00		0.01

6 - 11/07/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,470.00	980.00	6,800.00		7.90
DL	100.00	100.00	100.00		0.01

6 - 06/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,060.00	1,110.00	6,030.00		5.92
DL	100.00	100.00	100.00		0.01

7 - 11/04/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,840.00	930.00	5,960.00		6.60
DL	100.00	100.00	100.00		0.01

7 - 06/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,660.00	920.00	5,280.00		5.01
DL	100.00	100.00	100.00		0.01

8 - 11/07/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,950.00	1,050.00	7,210.00		7.90
DL	100.00	100.00	100.00		0.01

8 - 06/04/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,070.00	860.00	5,080.00		3.49
DL	100.00	100.00	100.00		0.01

9 - 11/05/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,810.00	1,220.00	8,320.00		7.80
DL	100.00	100.00	100.00		0.01

9 - 06/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,060.00	920.00	5,630.00		5.19
DL	100.00	100.00	100.00		0.01

Almonds - 10/24/2018: Almond Field Run

Almonds

Sample and source description: Almonds

Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 10.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,700.00	1,170.00	13,600.00		16.80
DL	100.00	100.00	100.00		0.01

L-1 - 11/05/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,580.00	1,080.00	7,500.00		9.20
DL	100.00	100.00	100.00		0.01

L-1 - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 49.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,220.00	1,770.00	10,100.00		7.01
DL	100.00	100.00	100.00		0.01

L-2 - 11/05/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,400.00	1,370.00	9,300.00		7.20
DL	100.00	100.00	100.00		0.01

L-2 - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,420.00	1,250.00	7,670.00		6.29
DL	100.00	100.00	100.00		0.01

L-3 - 11/06/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,220.00	880.00	5,730.00		7.00
DL	100.00	100.00	100.00		0.01

L-3 - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,930.00	770.00	5,210.00		5.31
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

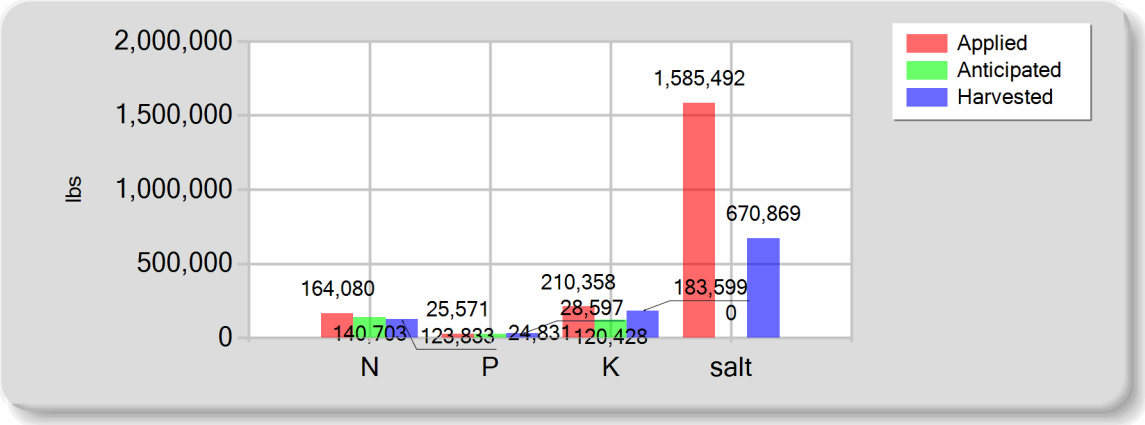
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

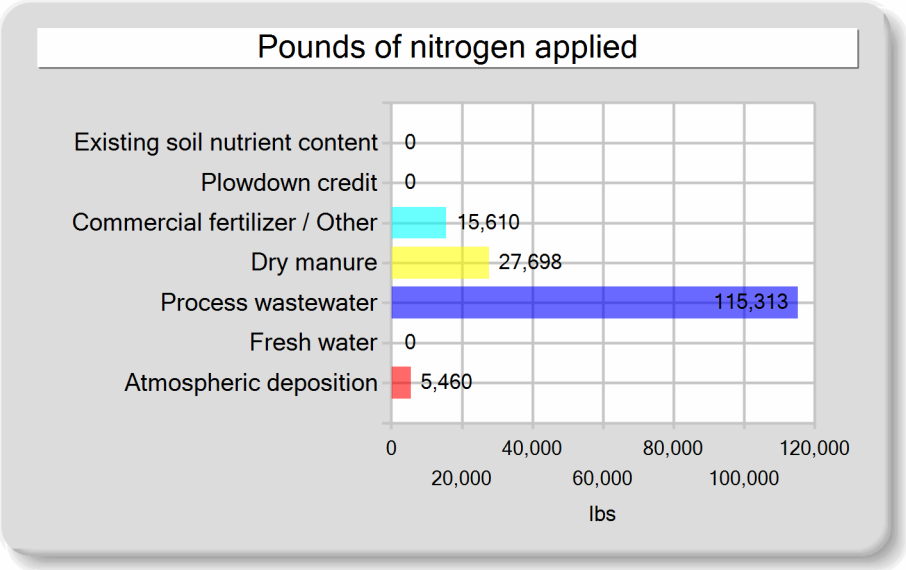
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

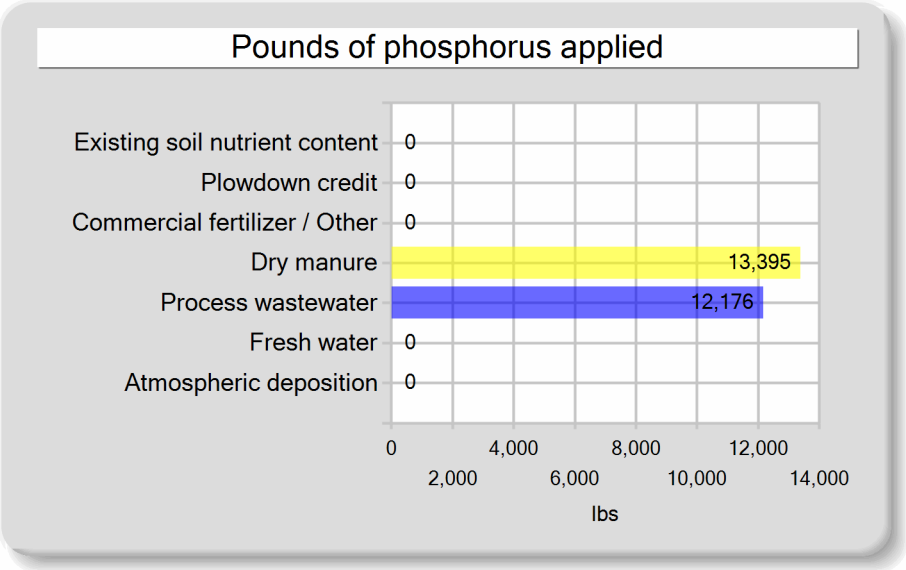
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	15,609.60	0.00	0.00	0.00
Dry manure	27,697.68	13,394.70	38,645.97	443,785.50
Process wastewater	115,313.01	12,176.43	171,711.96	1,017,224.79
Fresh water	0.00	0.00	0.00	124,481.80
Atmospheric deposition	5,460.00	0.00	0.00	0.00
Total nutrients applied	164,080.29	25,571.13	210,357.93	1,585,492.09
Anticipated crop nutrient removal	140,702.50	24,830.55	120,427.65	0.00
Actual crop nutrient removal	123,832.82	28,597.38	183,599.04	670,869.34
Nutrient balance	40,247.47	-3,026.25	26,758.89	914,622.74
Applied to removed ratio	1.33	0.89	1.15	2.36

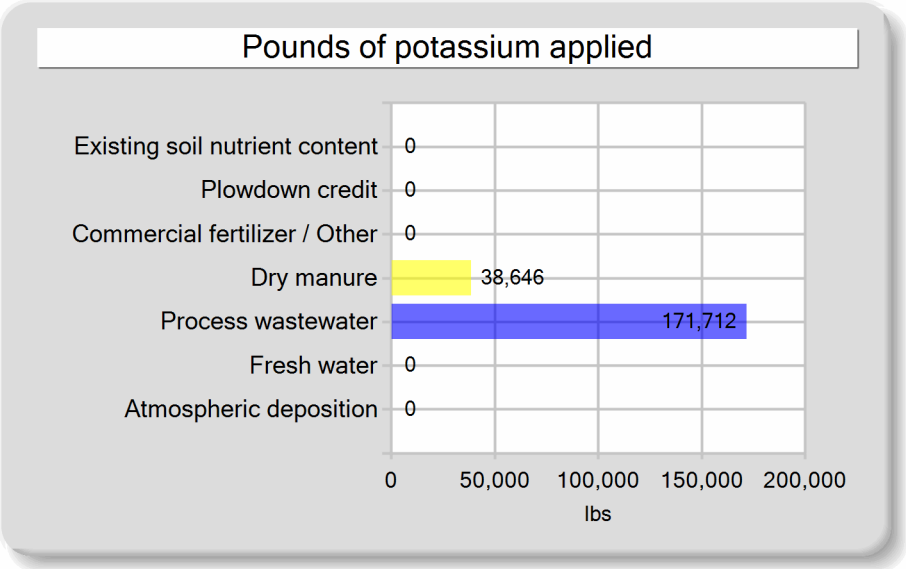
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

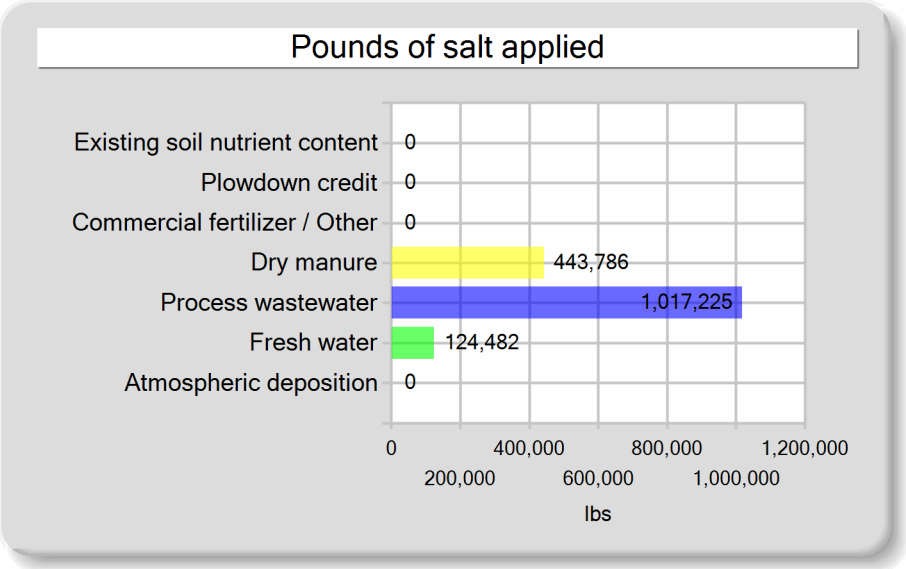


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Leonard A Pareira	Justin Pareira
SIGNATURE OF OWNER OF FACILITY	**SIGNATURE OF OPERATOR OF FACILITY**
Leonard Pareira	Justin Pareira
PRINT OR TYPE NAME	**PRINT OR TYPE NAME**
6-13-24	6-13-24
DATE	**DATE**

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Pareira Dairy
2487 W. Turlock Rd.
Snelling, CA 95369

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 3/2/23 4:08 PM
SUBMITTED BY: Sean
ANALYZED DATE: 3/3/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 21.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW #1	T3061003A	Electrical Conductivity (EC)	EPA 120.1 and	0.301	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.815	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	193	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	105	109 (T3061003A)	106 (T3061003A)	2.79
Electrical Conductivity (EC)	97.2			0.000
Nitrate Nitrogen (NO3-N)	101	82.5 (T3065001A)	80.6 (T3065001A)	2.38

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Snelling, CA 95369

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 3/2/23 4:08 PM
SUBMITTED BY: Sean
ANALYZED DATE: 3/3/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 21.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW #2 Milk Barn	T3061003B	Electrical Conductivity (EC)	EPA 120.1 and	0.168	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.46	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	108	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	105	109 (T3061003A)	106 (T3061003A)	2.79
Electrical Conductivity (EC)	97.2			0.000
Nitrate Nitrogen (NO3-N)	101	82.5 (T3065001A)	80.6 (T3065001A)	2.38

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Grower

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F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 3/2/23 4:08 PM
SUBMITTED BY: Sean
ANALYZED DATE: 3/3/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 21.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW #3 Heifers	T3061003C	Electrical Conductivity (EC)	EPA 120.1 and	0.450	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	9.77	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	288	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	105	109 (T3061003A)	106 (T3061003A)	2.79
Electrical Conductivity (EC)	97.2			0.000
Nitrate Nitrogen (NO3-N)	101	82.5 (T3065001A)	80.6 (T3065001A)	2.38

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ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.
865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: PAREIRA DAIRY
Mailing address:
(not facility address) SNELLING CA
Phone#:

Authorized Copy Released To: F&R AG SERVICES
Sampled By: Sean Locke
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	3/2/23	9:05	DW #1	DOMESTIC	21.2°		X				
2		9:10	DW #2 MILK BARN		21.2°		X				
3		9:15	DW #3 HEIFERS		21.2°		X				
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
		F&R AG SERVICES				
2nd		DENELE ANALYTICAL, INC.	3-2-23 4:08			T3061003

Required to be submitted to any agency? (YES) (NO)